School of Computing Sciences &Engineering

Department Of Computer Science & Engineering

Agentic Ai - Lab

(CSCR3215)

Lab File (2025-26)

For

B.Tech. (CSE) 6th Semester

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RAG Based Question Answering System

Problem Statement

The objective of this project is to build a Retrieval-Augmented Generation (RAG) system that answers questions based on a financial annual report (Apple Inc. 2025 10-K Report).

Instead of relying only on a language model's memory, the system retrieves relevant content from the uploaded PDF document and then generates an answer using that retrieved context.

This approach reduces hallucination and ensures factual accuracy.

Dataset / Knowledge Source

- **Dataset: Apple Inc. 2025 Form 10-K (PDF)**
- **Embedding Model: all-MiniLM-L6-v2**
- **Vector DB: ChromaDB**
- **Generator: Llama 3 (via Groq API)**

RAG Architecture

RAG Pipeline Steps:

1. Load PDF document
2. Chunk text
3. Generate embeddings
4. Store embeddings in ChromaDB
5. Retrieve relevant chunks
6. Pass retrieved context to LLM
7. Generate final answer

Simple Block Diagram

User Question
↓
Embedding Model
↓
Vector Database (ChromaDB)
↓
Top-K Retrieval
↓
LLM (Llama 3)
↓

Final Answer

Text Chunking Strategy

- Chunk size: 1000 characters
- Chunk overlap: 200 characters

  Reason:

- Prevents context loss
- Maintains semantic continuity
- Improves retrieval accuracy
- Avoids sentence truncation

Embedding Details

- **Model Used:** sentence-transformers/all-MiniLM-L6-v2
- Dimension: 384

  Reason:

- Lightweight
- Fast processing
- Works well on Colab
- Good semantic similarity performance

Vector Database

- Used: ChromaDB
- Storage Type: Persistent Local Storage

Reason:

- Fast similarity search
- Easy integration with LangChain
- Efficient semantic retrieval
- Supports persistent storage

Future Improvements

- Implement semantic chunking
- Hybrid search (BM25 + Vector Search)
- Add re-ranking layer
- Add metadata filtering
- Deploy full UI using Streamlit Cloud
- Upgrade to larger LLM models